

Login Credentials & Architecture Diagram

Login Credentials & Role Access

SupportLens uses a frictionless role-switching system for demonstration and judging purposes. No username or password is required to test the core functionality.

To switch between Agent and Customer roles during judging:

1. Open the application at `http://localhost:3000` (or your provided Demo URL).
2. On the landing page, you will see a prominent toggle: **"I am joining as: Agent / Customer"**.
3. **Agent Role:** Select "Agent", enter your name and a Room ID (e.g., 123). This grants you privileges to draw AR annotations.
4. **Customer Role:** Open a second browser window, select "Customer", enter a name and the **exact same Room ID** (e.g., 123).

You will instantly be connected in the same room with the correct role-based permissions applied.

Architecture Diagram

Our solution utilizes an Enterprise-Grade SFU Architecture. Rather than relying on simple peer-to-peer mesh WebRTC, we route all media through a C++ Mediasoup server. This ensures we strictly adhere to the rule of routing media through our own infrastructure while enabling features like graceful network degradation.

```
graph TD
    subgraph Frontend ["Next.js Web Client"]
        UI[User Interface - React]
        WC[WebRTC Client Device]
        SC[Socket.io Client]
        AR[AR Canvas Overlay]
        UI --> WC
        UI --> SC
        UI --> AR
    end
    subgraph Backend ["Node.js Server"]
        API[Express HTTP Server]
        SIG[Socket.io Signaling Server]
        subgraph SFU ["Mediasoup C++ Engine"]
            W[Worker Process]
            R[Router]
            T[WebRTC Transports]
            P[Producers - Receive Media]
            C[Consumers - Send Media]
            W --> R
            R --> T
            T --> P
            T --> C
        end
        API --> SIG
        SIG --> SFU
    end
    %% Connections
    WC <--> RTP[RTCP Media Streams]
    T <--> WS[WebSocket Signaling & Chat]
    SIG <--> AR[Draw Coordinates]
    classDef frontend fill:#3b82f6,stroke:#1d4ed8,stroke-width:2px,color:white;
    classDef backend fill:#10b981,stroke:#047857,stroke-width:2px,color:white;
    classDef engine fill:#8b5cf6,stroke:#6d28d9,stroke-width:2px,color:white;
    class UI,WC,SC,AR frontend;
    class API,SIG backend;
    class W,R,T,P,C engine;
```